

MASTER CAREER REPOSITORY & KNOWLEDGE BASE - PARIKSHIT KUMAR

--- SECTION 1: PROFESSIONAL SUMMARY & CAREER OBJECTIVE --- I am a versatile and highly adaptable software engineer with a passion for solving complex architectural bottlenecks. Over the years, I have explored the entire software development lifecycle, building everything from low-level game engine mechanics in C++ to interactive mobile applications using React Native. My core philosophy is to choose the right tool for the job, whether that involves deploying scalable backend microservices, optimizing front-end rendering paths, or training predictive machine learning models. I thrive in environments where I can wear multiple hats and continuously learn new paradigms.

--- SECTION 2: COMPREHENSIVE PROJECT PORTFOLIO (8 PROJECTS) ---

Project 1: "PixelQuest" - Custom 2D Game Engine

- **Tech Stack:** C++, OpenGL, SDL2, CMake.
- **Description:** Engineered a custom 2D game engine from scratch to understand computer graphics and memory management.
- **Details:** Implemented a custom entity-component-system (ECS) for efficient game object rendering. Wrote custom memory allocators to prevent fragmentation during gameplay. Handled matrix transformations and custom shader compilation using GLSL for dynamic lighting effects.

Project 2: Real-Time Fraud Detection Streaming Pipeline

- **Tech Stack:** Python, Apache Kafka, FastAPI, PostgreSQL, Docker.
- **Description:** Built a high-throughput event-streaming architecture designed to intercept and analyze simulated financial transactions for anomalies in real-time.
- **Details:** Configured an Apache Kafka cluster utilizing Zookeeper to handle an ingestion rate of 10,000 messages per second. Developed a Python-based consumer microservice that processed Kafka topics, executed rule-based anomaly detection algorithms, and asynchronously stored flagged transactions in a heavily indexed PostgreSQL database. Containerized the entire ingestion pipeline using Docker Compose for seamless local deployment.

Project 3: "Nyaya Setu" - Offline Legal Advisor App

- **Tech Stack:** Java, Spring Boot, SQLite, Thymeleaf.
- **Description:** A localized, offline-first application providing rural citizens access to complex legal documentation without internet dependency.

- **Details:** Designed a heavily compressed local SQLite database schema to store 50,000+ legal statutes. Built a Java Spring Boot backend to quickly query and render legal clauses using Thymeleaf templates. Explicitly avoided specialized UI tools like AWT or paid cloud systems to ensure the software remained entirely free and accessible on low-end hardware.

Project 4: "FitTrack" - Cross-Platform Mobile Fitness Tracker

- **Tech Stack:** JavaScript, React Native, Redux, Firebase.
- **Description:** Developed an interactive iOS and Android application for users to track macronutrients, log gym sessions, and visualize weightlifting progress.
- **Details:** Designed responsive mobile UI components. Handled global application state using Redux to ensure immediate UI updates when users logged a workout. Integrated Firebase for remote user authentication and syncing user profiles across multiple devices.

Project 5: Open-Source Kubernetes Cluster Orchestrator

- **Tech Stack:** Golang, Kubernetes (K8s), Docker, gRPC, Prometheus.
- **Description:** Engineered a custom command-line interface (CLI) tool in Go to automate the deployment, scaling, and monitoring of localized microservices across a Kubernetes cluster.
- **Details:** Utilized Golang's concurrency model (Goroutines) to execute simultaneous health checks across dozens of active Docker containers. Wrote custom Kubernetes deployment manifests (YAML) to manage replica sets and load balancers. Implemented gRPC protocols for lightning-fast internal communication between microservices, bypassing standard REST latency.

Project 6: E-Commerce Inventory Management Monolith

- **Tech Stack:** PHP, Laravel, MySQL, Bootstrap.
- **Description:** A standard web application built for a local retail shop to track their warehouse stock, generate invoices, and handle basic employee payroll.
- **Details:** Designed a monolithic MVC architecture using Laravel. Built robust relational tables in MySQL to handle complex joins between products, suppliers, and customer orders. Created an administrative dashboard using Bootstrap CSS for local store managers to easily update stock quantities.

Project 7: High-Frequency Distributed Caching Layer

- **Tech Stack:** Golang, Redis, Linux, Bash Scripting.
- **Description:** Created a custom middleware caching proxy to drastically reduce database load for read-heavy application architectures.
- **Details:** Leveraged Golang to build a lightning-fast proxy server that intercepted API requests. Integrated a localized Redis cluster to cache complex query results, utilizing Redis TTL (Time-To-Live) commands to automatically expire stale data. Wrote bash scripts to automate the spin-up and configuration of the Redis nodes on local Linux virtual machines.

Project 8: Natural Language Processing (NLP) Sentiment Analyzer

- **Tech Stack:** Python, Pandas, Scikit-Learn, NLTK.
- **Description:** A machine learning pipeline designed to read thousands of product reviews and automatically categorize them as positive, negative, or neutral.
- **Details:** Cleaned and pre-processed 100,000+ text data points using Pandas and NLTK. Trained a Support Vector Machine (SVM) and a Naive Bayes classifier using Scikit-Learn. Achieved an 88% accuracy rate on the validation dataset.

--- SECTION 3: SKILLS DUMP --- Programming: Java, Python, C++, Golang, JavaScript, PHP. Databases: PostgreSQL, MySQL, SQLite, Redis, Firebase. DevOps/Infrastructure: Kubernetes, Docker, Apache Kafka, Linux, Git, Jenkins, GitHub Actions. Frameworks: Spring Boot, FastAPI, React Native, Laravel.